Optimization impact vs streaming baseline

Updated 2026-10-13T06:56:35+02:00. Campaigns terminated.

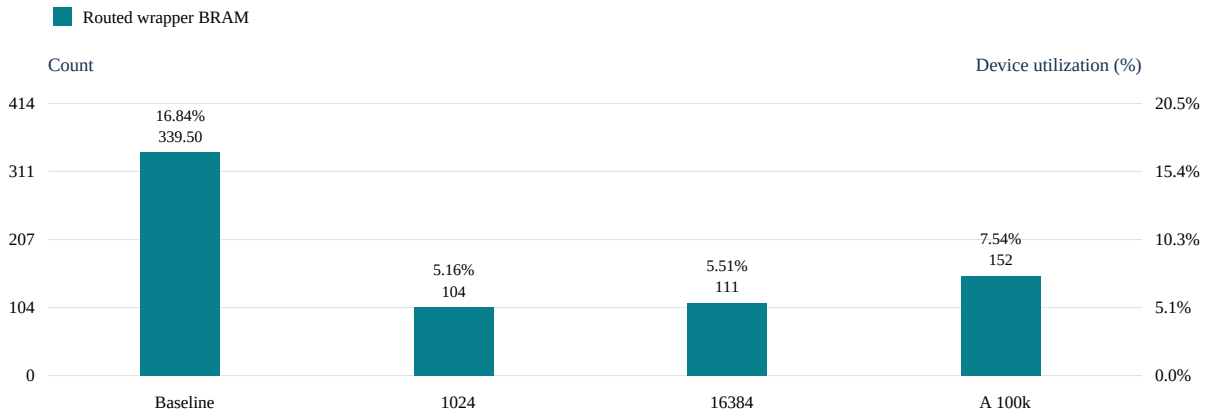
Two references: R_S is the existing routed, board-tested streaming design. OPT_1_STREAMING regenerates the same intended model and preprocessing with the D_I interface and timestamps, matching the new candidates. Its routed result remains separate. No custom conv0 kernel is used.

Interpretation: reductions versus R_S are whole-variant comparisons, including integration differences. Reductions versus OPT_1_STREAMING isolate the configured optimization with matching instrumentation. Savings % uses the reference count; device utilization % uses U55C capacity.

1. A and FIFO sweeps: BRAM savings

Variant	BRAM tiles	Device %	Saved tiles vs OPT_1	Saved % vs OPT_1
Streaming baseline	339.5	16.84%	0	0.00%
FIFO 1024	104	5.16%	235.5	69.37%
FIFO 16384	111	5.51%	228.5	67.30%
A / 100000	152	7.54%	187.5	55.23%

BRAM usage: matching baseline and FIFO variants



Left: BRAM tiles. Right: % of device BRAM. Bar labels show both.

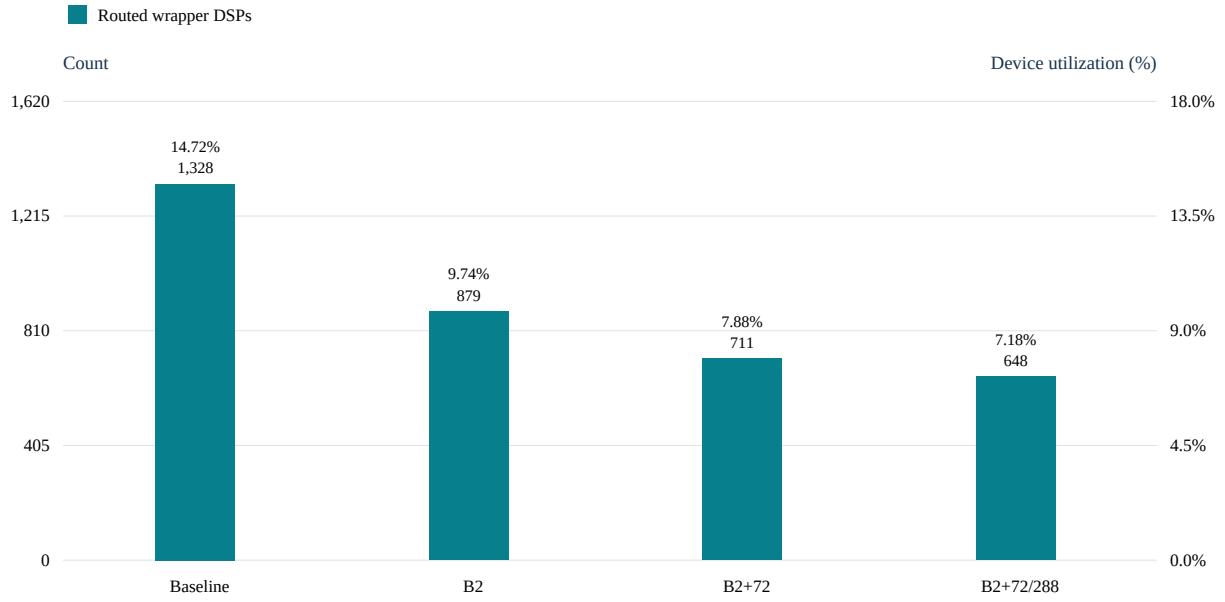
A is the 100000-depth sweep point. Profiling depth is the temporary profiling capacity, not the depth of every final FIFO. Logical FIFO storage is not substituted for routed BRAM usage.

2. B series: DSP savings

Variant	DSPs	Device %	Saved DSPs vs OPT_1	Saved % vs OPT_1
Streaming baseline	1,328	14.72%	0	0.00%
B2	879	9.74%	449	33.81%
B2 + conv3/4 RF72	711	7.88%	617	46.46%
B2 + RF72/288	648	7.18%	680	51.20%

Reuse factors for conv3 / conv4 / conv5 / conv6: baseline and R_S = 36 / 36 / 32 / 32; B2 = 36 / 36 / 144 / 144; B2+72 = 72 / 72 / 144 / 144; aggressive B2+72/288 = 72 / 72 / 288 / 288.

DSP usage: matching baseline and B variants



Left: DSP count. Right: % of 9024 device DSPs. Bar labels show both.

B2 shares more multipliers in conv5/6. B2+72 also increases reuse in conv3/4. All B variants retain the baseline FIFO configuration; no B variant includes FIFO optimization A.

Higher reuse can increase latency or reduce throughput. Resource savings and latency must be judged together. B3 is a separate proposal and is not included as a measured variant.

Streaming vs non-streaming baseline

OPT_0_NONSTREAMING uses the same stock hls4ml CNN, weights, quantizers, reuse settings and ABI v3 timestamps as OPT_1_STREAMING. Its synchronous 65536 x 16-bit buffer waits for both the complete preprocessed image and the last raw input beat before the CNN consumes a pixel. No cross-frame overlap is enabled.

The sequencing mirrors the original production preprocessing-then-CNN design. The split wrappers and timestamps match this campaign, so this is a matched baseline rather than a bit-identical replay of the original production package.

Variant	BRAM tiles	Device %	DSPs	Device %
Non-streaming baseline	369.5	18.33%	1,328	14.72%
Streaming baseline	339.5	16.84%	1,328	14.72%

Standalone buffer synthesis inferred 30 BRAM tiles (1.49% of U55C BRAM). This is a component estimate, not the complete routed baseline cost. The full build remains pending until all parity, actual-build RTL and timing gates pass.

Timestamp scopes for later benchmarking: raw first-to-last measures input arrival; first-to-last preprocessing output measures preprocessing delivery; first CNN input-to-score measures the observed CNN interval; arm-to-verdict includes the full frame fill. Host file reading remains outside FPGA timestamps and requires host-side measurement.

Variant	RTL arm-to-verdict (ms)	Production e2e
Non-streaming baseline	1.366616	not measured
Streaming baseline	pending	not measured

FIFO depths 4096, 65536 and 262144 are no longer selected for routing. Their completed native profiles remain archived. No profiling is repeated, and no board benchmarks are launched.

3a. R_S: measured host latency

R_S has existing U55C board measurements. These are retained as a historical reference, separate from the RTL simulations of the current optimization variants.

R_S input	Bytes	Median (ms)	p95 (ms)
synthetic	4,194,304	1.105846	1.106645
real	5,540,996	1.105871	1.107379
real	12,004,440	1.105781	1.106776
synthetic	16,777,216	1.344188	1.350839
real	18,876,484	1.508383	1.509619
real	30,323,588	2.403948	2.413747
real	34,162,708	2.704476	2.717845
real	41,278,872	3.261584	3.379955
real	48,540,364	3.829969	3.948390
real	55,676,772	4.388527	4.508459
synthetic	67,108,864	5.283494	5.402412
synthetic	134,217,664	10.536761	10.774956

Recorded scope: host start to host-observed result. Includes submission/flush, transfer of the complete input, FPGA streaming preprocessing and inference, result DMA and readback. Each case has 20 warm-ups and 200 measured calls.

Excluded: file reading, input copies, allocation, registration and FPGA programming. Thus even this historical result is not full file-read-to-verdict latency. R_S has no internal FPGA timestamp breakdown.

At 4 MiB the median is 1.1058465 ms. This is not directly comparable to the approximately 1.1 ms first-CNN-input-to-result RTL interval below: endpoints, workload sizes, transport and execution environments differ. No speedup is computed between these scopes.

OPT_1_STREAMING and the A/FIFO/B variants have no board latency measurements in this campaign. Their host/e2e comparison remains unmeasured; the current request does not launch benchmarks.

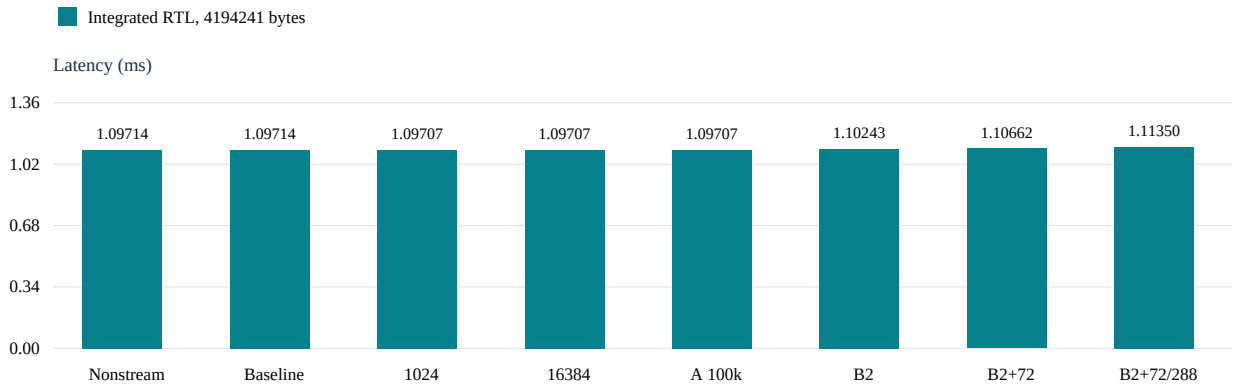
3b. Latency across current variants

Production end-to-end latency is unavailable for the eight selected campaign variants. No new board benchmarks were run; historical R_S timings are listed separately on the preceding page. Host bitstream reading, host preprocessing if applicable, DMA, FPGA preprocessing, inference and result return have not been timed together.

The available comparison below is **integrated RTL: first CNN input to accepted result**. It includes waiting for later pixels from the streaming preprocessor, but excludes host I/O and the interval before the first CNN input. For non-streaming this interval excludes the entire frame fill. It must not be labelled production e2e latency.

Variant	Production e2e	RTL 4,194,241 B (ms)	RTL 4,194,240 B (ms)	RTL change vs base
Non-streaming baseline	not measured	1.097140	1.097140	+0.000%
Streaming baseline	not measured	1.097140	16.800848	+0.000%
FIFO 1024	not measured	1.097068	16.800776	-0.007%
FIFO 16384	not measured	1.097068	16.800776	-0.007%
A / 100000	not measured	1.097068	16.800776	-0.007%
B2	not measured	1.102428	16.806136	+0.482%
B2 + conv3/4 RF72	not measured	1.106624	16.810332	+0.864%
B2 + RF72/288	not measured	1.113504	16.817212	+1.492%

Available latency comparison (not production e2e)



Milliseconds; same input size, testbench stalls, and output backpressure.

The 4,194,240-byte case uses a slower bytewise preprocessing path; 4,194,241 bytes uses the fast path. Keep these cases separate. All comparisons use 4 ns clocks and identical synthetic inputs. The earlier baseline validation is reused only after checking matching model, HLS, interface and validation configuration, testbench, and sealed validation evidence.

Appendix A. All resource types

The eight selected configurations plus the historical R_S reference. Routed user-wrapper counts include checker integration and debug overhead; full-shell overhead is excluded. A is listed once as the 100000-depth point. Pending is not zero.

Variant	LUT	FF	BRAM tiles	DSP	URAM
R_S (measured reference)	63,764	82,210	345	1,328	11
Non-streaming baseline	68,269	87,137	369.5	1,328	11
Streaming baseline	68,180	87,042	339.5	1,328	11
FIFO 1024	65,683	91,729	104	1,328	0
FIFO 16384	66,148	92,641	111	1,328	0
A / 100000	65,996	93,477	152	1,328	0
B2	64,552	84,545	322.5	879	11
B2 + conv3/4 RF72	64,816	82,820	309	711	11
B2 + RF72/288	63,231	81,645	290	648	11

Device utilization percentages, same wrapper scope

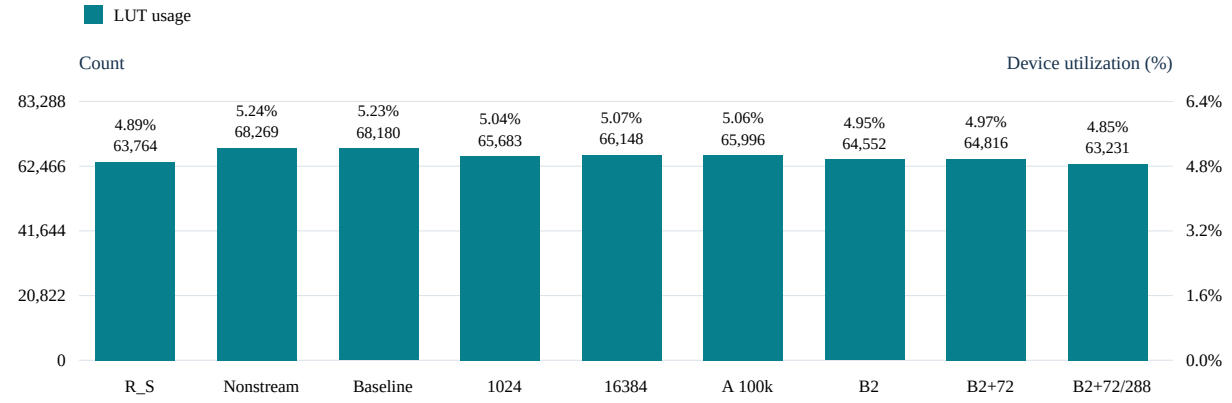
Variant	LUT	FF	BRAM tiles	DSP	URAM
R_S (measured reference)	4.89%	3.15%	17.11%	14.72%	1.15%
Non-streaming baseline	5.24%	3.34%	18.33%	14.72%	1.15%
Streaming baseline	5.23%	3.34%	16.84%	14.72%	1.15%
FIFO 1024	5.04%	3.52%	5.16%	14.72%	0.00%
FIFO 16384	5.07%	3.55%	5.51%	14.72%	0.00%
A / 100000	5.06%	3.59%	7.54%	14.72%	0.00%
B2	4.95%	3.24%	16.00%	9.74%	1.15%
B2 + conv3/4 RF72	4.97%	3.18%	15.33%	7.88%	1.15%
B2 + RF72/288	4.85%	3.13%	14.38%	7.18%	1.15%

Device capacity	LUT	FF	BRAM tiles	DSP	URAM
U55C	1,303,680	2,607,360	2,016	9,024	960

BRAM tiles = RAMB36 + RAMB18 / 2. Percentages use the full device capacity, not a placement region or an optimization budget.

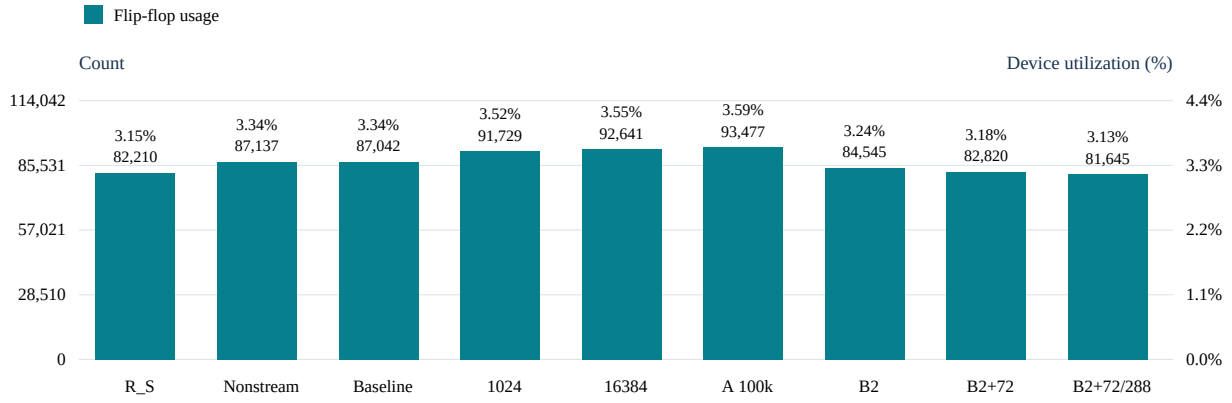
Appendix B. Resource plots

LUT usage across all variants



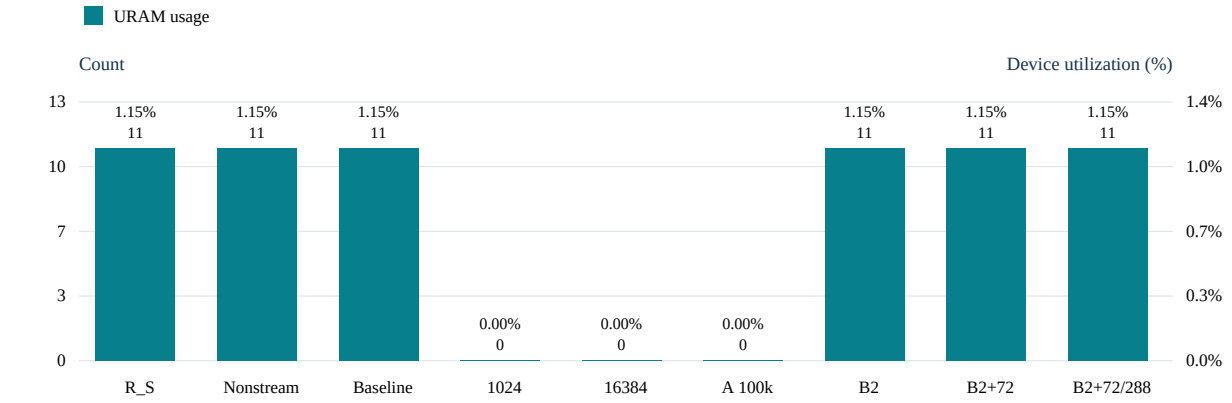
Left: count. Right: device utilization %. Bar labels show both.

Flip-flop usage across all variants



Left: count. Right: device utilization %. Bar labels show both.

URAM usage across all variants



Left: count. Right: device utilization %. Bar labels show both.

BRAM and DSP plots for their optimization families appear in sections 1 and 2. Appendix A supplies all resource counts and percentages across both families.

Appendix C. Full design and evidence

Full routed design, including the Coyote shell: each cell gives count / device utilization. Use the user-wrapper tables to attribute optimization impact; shell overhead can mask it.

Variant	LUT	FF	BRAM tiles	DSP	URAM
R_S (measured reference)	184,129 / 14.12%	253,001 / 9.70%	487 / 24.16%	1,328 / 14.72%	11 / 1.15%
Non-streaming baseline	189,158 / 14.51%	259,187 / 9.94%	511.5 / 25.37%	1,328 / 14.72%	11 / 1.15%
Streaming baseline	188,993 / 14.50%	258,992 / 9.93%	481.5 / 23.88%	1,328 / 14.72%	11 / 1.15%
FIFO 1024	186,350 / 14.29%	263,665 / 10.11%	246 / 12.20%	1,328 / 14.72%	0 / 0.00%
FIFO 16384	187,000 / 14.34%	264,608 / 10.15%	253 / 12.55%	1,328 / 14.72%	0 / 0.00%
A / 100000	186,868 / 14.33%	265,459 / 10.18%	294 / 14.58%	1,328 / 14.72%	0 / 0.00%
B2	185,282 / 14.21%	256,481 / 9.84%	464.5 / 23.04%	879 / 9.74%	11 / 1.15%
B2 + conv3/4 RF72	185,566 / 14.23%	254,740 / 9.77%	451 / 22.37%	711 / 7.88%	11 / 1.15%
B2 + RF72/288	184,020 / 14.12%	253,613 / 9.73%	432 / 21.43%	648 / 7.18%	11 / 1.15%

Build state

Variant	Stage
R_S (measured reference)	Historical routed build; board-tested; different interface
Non-streaming baseline	Complete; correctness and timing passed
Streaming baseline	Complete; correctness and timing passed
FIFO 1024	Complete; correctness and timing passed
FIFO 16384	Complete; correctness and timing passed
A / 100000	Complete; correctness and timing passed
B2	Complete; correctness and timing passed
B2 + conv3/4 RF72	Complete; correctness and timing passed
B2 + RF72/288	Complete; correctness and timing passed

Source paths and hashes are recorded in streaming_optimization_results_20261012.json. R_S uses the archived u55c02_20261009 report, verified against its file manifest and original routed reports. Its interface is a single HLS wrapper with production DMA, not the split timestamped integration of OPT_1_STREAMING. Routed results are checked against sealed artifacts, source identity, correctness and timing gates. Integrated RTL measurements come from rtl_validation/result.json. OPT_1_STREAMING latency currently uses the matching earlier attempt_02 when the fresh baseline has not yet validated.

The build-host reboot at 07:00 CEST on 12 October interrupted builds and local monitors. They were restarted at 15:58 after verification. The report watcher refreshes this PDF independently when the primary queue and the non-streaming queue terminate. No board benchmarking is scheduled.